# JAVA SWING

**Introduction to Java Swing**

Java Swing is a part of Java used to develop **Graphical User Interface (GUI)** applications.

It is included in **Java Foundation Classes (JFC)** and provides a rich set of components to build modern desktop applications.

The term GUI means:

- Interaction using **visual elements**
- Instead of typing commands

Swing applications are:

✔ Platform Independent

✔ Lightweight

✔ Flexible

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## 🔷 What is GUI?

GUI (Graphical User Interface) allows users to interact with the system using:

- Windows
- Buttons
- Text Fields
- Menus

Example:

- ATM Machine
- Login Form
- Calculator

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## ◆ Basic Concepts of Java Swing

A Java Swing program is mainly based on:

1. Frame (Container)
2. Layout Manager (Arrangement)
3. Components (UI Elements)

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## ◆ 1. Frame (JFrame)

A **Frame** is the top-level container used to create a window.

✓ It acts as the main screen

✓ All components are added inside it

### Syntax:

```
JFrame frame = new JFrame("Title");
```

**Important Methods:**

- setSize(width, height) → Set window size
- setVisible(true) → Display frame
- setDefaultCloseOperation() → Close operation

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## ◆ 2. Layout Manager

A Layout Manager controls how components are arranged inside a container.

Without layout, components may overlap.

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**Types of Layout Managers**

### 1. FlowLayout

- Arranges components from **left to right**
- Default layout in JFrame

frame.setLayout(new FlowLayout());

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### 2. BorderLayout

- Divides frame into **5 regions**:
  - North
  - South
  - East
  - West
  - Center

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### 3. GridLayout

- Arranges components in **rows and columns**

```
frame.setLayout(new GridLayout(2,2));
```

### Components in Swing

Components are GUI elements added to the frame.

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### Common Components

### JLabel

- Used to display text

```
JLabel label = new JLabel("Hello");
```

**JTextField**

- Used to take user input

JTextField text = new JTextField(10);

```
JTextField text = new JTextField(10);
```

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**JButton**

- Used to perform actions

JButton btn = new JButton("Submit");

```
JButton btn = new JButton("Submit");
```

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**JCheckBox**

- Used for selection

JCheckBox cb = new JCheckBox("Accept");

```
JCheckBox cb = new JCheckBox("Accept");
```

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## Steps to Create a Swing Program

1. Create JFrame
2. Set Size of Frame
3. Set Layout
4. Create Components
5. Add Components to Frame
6. Set Visibility

```java
import javax.swing.*;
import java.awt.*;

public class SwingProgram {
    public static void main(String[] args) {

        // Step 1: Create Frame
        JFrame frame = new JFrame("Java Swing Example");

        // Step 2: Set Size
        frame.setSize(400, 300);

        // Step 3: Close Operation
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Step 4: Set Layout
        frame.setLayout(new FlowLayout());

        // Step 5: Create Components
        JLabel label = new JLabel("Enter Name:");
        JTextField textField = new JTextField(15);
        JButton button = new JButton("Submit");

        // Step 6: Add Components
        frame.add(label);
        frame.add(textField);
        frame.add(button);

        // Step 7: Make Visible
        frame.setVisible(true);
    }
}
```

JFrame → Creates main window

setSize() → Defines window size

FlowLayout → Arranges components in a row

JLabel → Displays text

JTextField → Accepts input

JButton → Used for user action

add() → Adds components to frame

setVisible(true) → Shows window

**Output**
A window appears with:

- Label → "Enter Name"
- Text Field
- Submit Button

**Advantages of Java Swing**
✓ Platform Independent
✓ Rich GUI Components
✓ Easy to use
✓ Customizable

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### 🔷 Disadvantages of Java Swing

1. Slower performance
2. Outdated look compared to JavaFX
3. More coding required

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### 🔷 Real Life Applications

- Banking Systems
- ATM Interfaces
- Desktop Applications
- Management Systems